

Quantum Data Loading

- Consider some nonunitary matrix $L \in \mathbb{C}^{N \times N}$. Ideally, we would decompose it as

$$L = \sum_{l=1}^{N_s} c_l L_l$$

where $L_l \in \mathbb{C}^{N \times N}$ are unitary matrices, $c_l \in \mathbb{C}$ and $N_s = \text{poly}(\log N)$. May be difficult or impossible

- By building upon GS2024, we developed a new approach to efficiently load sparse structured matrices.
- First, define the set $\mathbb{P} = \{\rho_0, \rho_1, \rho_2, \rho_3, \rho_4\}$ where

$$\rho_0 = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \quad \rho_1 = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \quad \rho_2 = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}, \quad \rho_3 = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}, \quad \rho_4 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

- We are interested in decompositions of the form

$$L = \sum_{l=1}^{N_s} c_l A_l$$

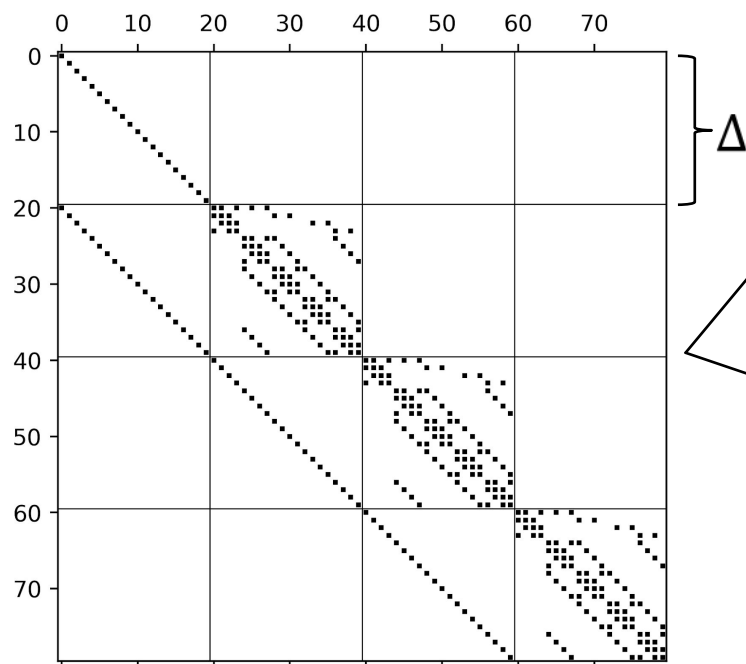
where $A_l \in \mathbb{C}^{N \times N}$ are nonunitary matrices of the form

$$A_l = \bigotimes_k \rho_{r_k}$$

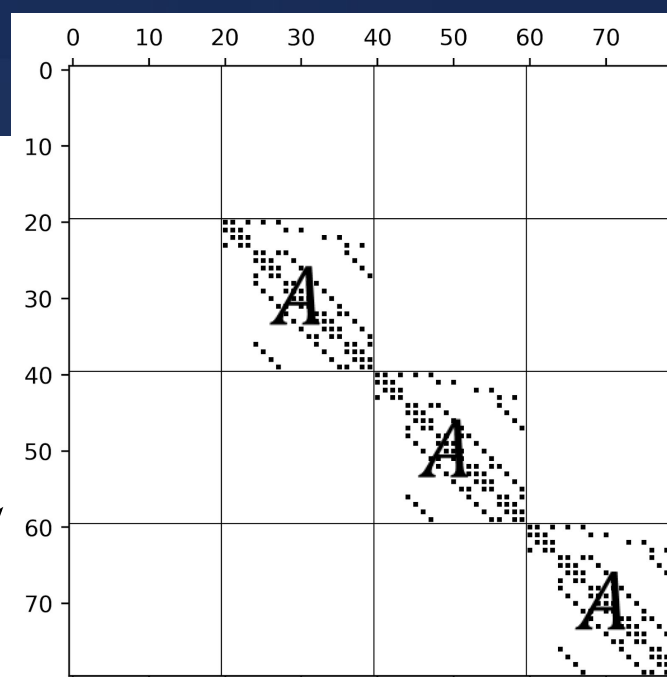
Example

Carleman Linearized System

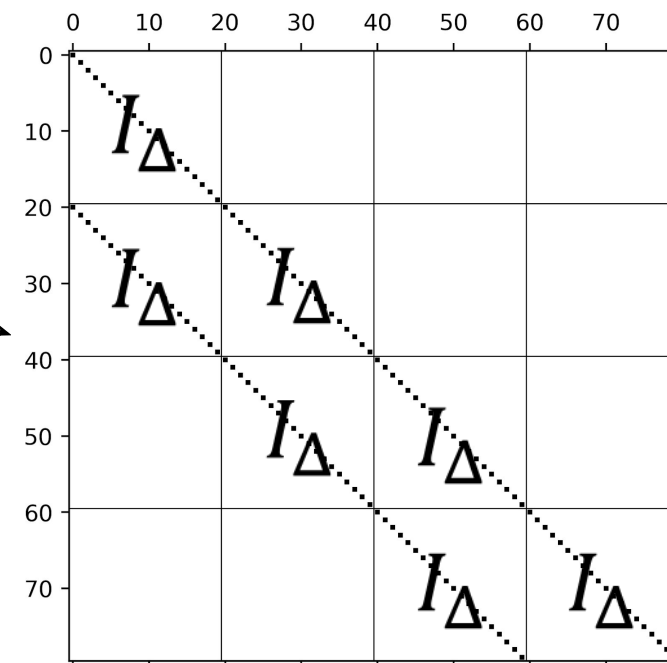
$$\begin{pmatrix} I & 0 & \dots & 0 \\ -I & (I - \Delta t A) & \dots & 0 \\ \vdots & \ddots & \ddots & \vdots \\ 0 & \dots & -I & (I - \Delta t A) \end{pmatrix} \begin{pmatrix} \hat{u}^1 \\ \hat{u}^2 \\ \vdots \\ \hat{u}^{n_t} \end{pmatrix} = \begin{pmatrix} \vec{u}(t=0) \\ 0 \\ \vdots \\ 0 \end{pmatrix}$$



$n_t = 4, \quad n_x = 4, \quad \alpha = 2$



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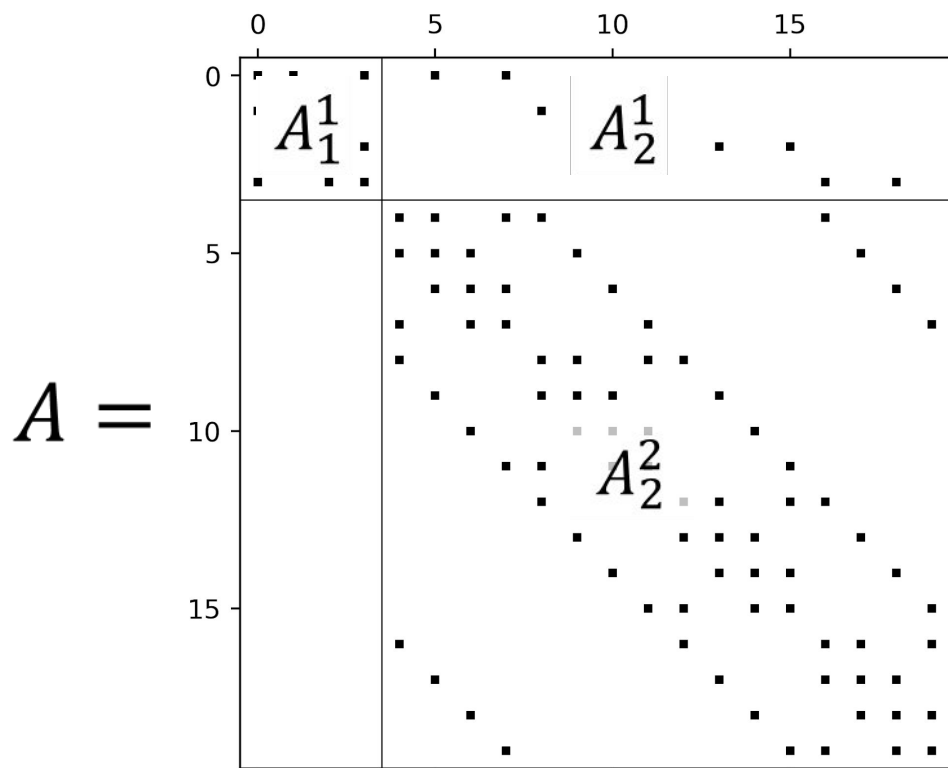
$$= (I^{\otimes 2} - \rho_0^{\otimes 2}) \otimes A$$

Needs further
decomposition

$$= I^{\otimes 2} \otimes I_{\Delta} \checkmark \\ - I \otimes \rho_2 \otimes I_{\Delta} \\ - \rho_2 \otimes \rho_1 \otimes I_{\Delta}$$

No further
decomposition

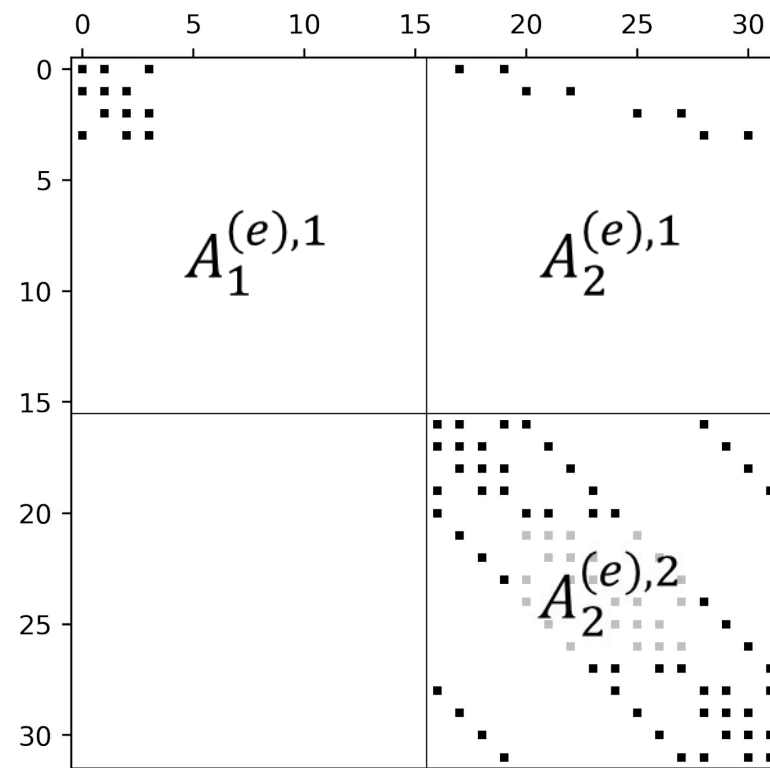
Example Continued



$$= \sum_j ?_j$$

Embed A
into $A^{(e)}$

$$A^{(e)} =$$



$$= \rho_0 \otimes A_1^{(e),1} \\ + \rho_1 \otimes A_2^{(e),1} \\ + \rho_3 \otimes A_2^{(e),2}$$